

KYIV, Ukraine

WMO: 333450

Lat: 50.3917N

Lon: 30.5356E

Elev: 548

StdP: 14.41

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.5	6.5	-5.5	4.2	3.6	-0.2	5.6	7.6	22.6	32.4	20.4	26.4	6.6	30	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	88.1	68.5	85.3	67.7	82.3	66.3	71.1	82.7	69.7	80.8	68.3	79.0	6.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	103.9	75.9	66.1	98.1	74.4	64.4	92.5	73.3	35.3	83.3	34.0	80.9	32.9	79.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.2	16.7	14.6	-4.1	92.5	6.2	3.8	-8.6	95.2	-12.2	97.4	-15.7	99.5	-20.3	102.3		
			-4.7	74.3	6.2	1.9	-9.1	75.6	-12.7	76.7	-16.2	77.8	-20.7	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	25.3	27.5	36.1	50.0	60.3	66.5	70.1	68.4	58.5	47.1	37.3	28.6
	DBStd	17.88	9.81	10.09	8.12	7.61	7.61	6.26	5.53	6.25	6.94	7.66	8.52	9.49
	HDD50	3142	767	631	433	92	10	0	0	0	11	148	388	662
	HDD65	6652	1232	1051	896	451	184	58	17	34	214	556	832	1127
	CDD50	2442	0	0	2	93	331	495	622	569	267	57	6	0
	CDD65	479	0	0	0	2	40	103	174	139	20	1	0	0
	CDH74	4066	0	0	0	21	342	838	1429	1243	189	4	0	0
	CDH80	1159	0	0	0	3	63	203	433	422	35	0	0	0
Wind	WSAvg	6.9	7.8	7.7	8.1	7.2	6.7	6.6	6.1	5.8	6.1	6.4	7.2	7.7
Precipitation	PrecAvg	24.8	1.8	1.6	1.5	1.7	2.3	2.9	3.1	2.4	2.1	1.6	2.0	2.0
	PrecMax	35.4	5.9	4.8	4.4	5.2	6.8	7.5	8.5	6.5	8.3	5.7	5.6	5.2
	PrecMin	15.0	0.3	0.2	0.1	0.1	0.5	0.3	0.2	0.1	0.1	0.0	0.2	0.1
	PrecStd	4.4	1.1	1.0	0.9	1.2	1.3	1.6	1.7	1.3	1.7	1.1	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	49.9	61.5	76.9	85.7	88.2	91.7	94.2	84.3	73.1	60.8	50.2
		MCWB	41.8	43.8	49.0	59.1	65.3	68.0	70.8	69.1	64.6	59.4	53.4	47.6
	2%	DB	42.1	45.0	56.9	71.1	81.0	84.9	87.9	89.2	79.0	67.7	55.4	45.6
		MCWB	39.5	40.8	46.5	55.0	63.0	66.9	69.4	68.7	62.3	58.3	51.4	43.6
	5%	DB	39.3	41.7	52.0	66.6	77.7	82.3	84.6	84.5	75.1	62.7	51.7	42.9
		MCWB	37.3	38.7	44.1	53.2	62.0	66.4	68.8	67.3	61.2	55.5	48.8	41.5
	10%	DB	36.6	39.0	48.1	63.0	74.3	79.0	82.0	80.7	70.8	58.8	48.5	39.7
		MCWB	35.1	36.4	42.1	51.1	60.5	65.1	67.4	65.7	59.6	53.4	46.5	38.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	44.9	50.7	60.6	68.0	71.6	73.6	73.2	67.6	62.6	55.2	47.9
		MCDB	44.0	49.0	59.6	74.1	80.0	82.0	85.9	87.0	80.0	69.6	58.9	49.8
	2%	WB	39.8	41.4	47.6	57.2	65.6	69.7	71.7	70.5	64.5	59.3	52.0	43.7
		MCDB	41.5	44.0	54.3	67.0	76.7	80.4	83.4	82.7	74.9	65.3	54.6	44.8
	5%	WB	37.3	39.0	44.9	54.8	63.7	68.0	70.1	68.8	62.4	56.6	49.3	41.3
		MCDB	38.8	41.5	50.3	63.6	74.0	78.6	81.1	80.6	71.8	62.1	51.5	42.5
	10%	WB	35.2	36.6	42.4	52.8	61.9	66.5	68.8	67.2	60.6	53.8	46.6	38.4
		MCDB	36.4	38.3	47.3	61.4	71.8	76.4	79.1	78.4	68.9	58.2	48.4	39.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.9	12.9	16.6	18.4	17.6	17.8	18.3	16.7	14.3	8.6	7.8
		MCDBR	8.4	12.2	20.1	22.5	22.8	21.9	22.1	24.2	24.3	21.2	12.7	9.9
		MCWBR	7.2	9.1	12.2	11.2	9.8	9.5	9.0	9.5	11.0	12.7	9.4	8.2
	5% WB	MCDBR	7.9	11.0	18.0	19.4	20.3	19.4	19.6	20.6	20.4	18.9	11.0	9.3
		MCWBR	7.1	8.8	11.9	10.6	9.6	9.5	8.8	9.2	10.5	12.2	8.9	8.1
Clear-Sky Solar Irradiance	taub	0.307	0.337	0.368	0.432	0.406	0.398	0.427	0.429	0.397	0.357	0.339	0.308	
	taud	2.226	2.176	2.233	2.137	2.289	2.330	2.261	2.242	2.296	2.385	2.324	2.267	
	Ebn at Noon	224	246	262	256	270	273	262	255	250	240	211	205	
	Edh at Noon	23	31	36	44	40	38	41	39	34	26	21	19	
All-Sky Solar Radiation	RadAvg	274	536	944	1376	1801	1902	1848	1609	1097	628	250	185	
	RadStd	49	92	150	140	141	139	121	99	163	100	38	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.12	N/A	N/A	+1.94	N/A	N/A	N/A	-234	+137	N/A
(47)	Regional (10 neighbors)	+1.33	N/A	N/A	+1.80	+0.90	N/A	-252	-362	+216	+99

Nomenclature: See separate page